Name: _____ Register No: _____

Experiment 3.a

Question: Conduct an experiment to show data visualization using line plot.

Code

```
import json
import re
from collections import Counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

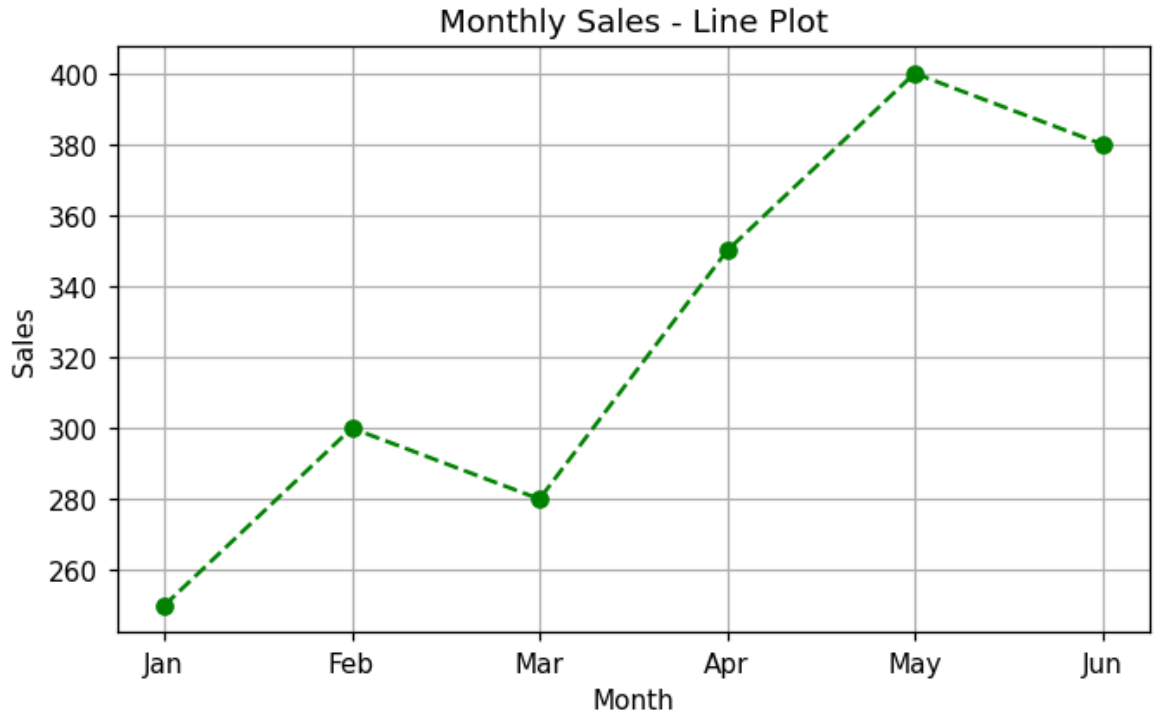
def exp_3a():
    months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
    sales = [250, 300, 280, 350, 400, 380]
    plt.figure(figsize=(7, 4))
    plt.plot(months, sales, marker="o", color="green", linestyle="--")
    plt.title("Monthly Sales - Line Plot")
    plt.xlabel("Month")
    plt.ylabel("Sales")
    plt.grid(True)
    plt.show()


exp_3a()
```

Output

```
(see graph below)
```

Graph



Experiment 3.b

Question: Conduct an experiment to show data visualization using bar chart.

Code

```
import json
import re
from collections import Counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

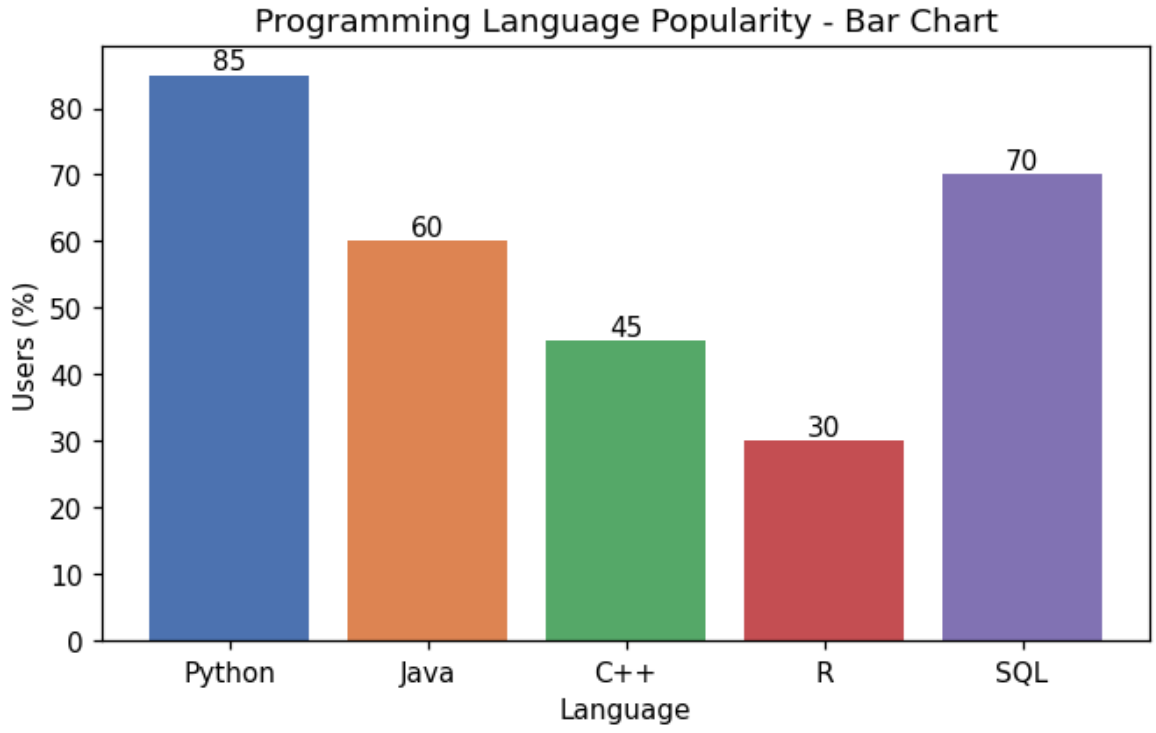
def exp_3b():
    langs = ["Python", "Java", "C++", "R", "SQL"]
    users = [85, 60, 45, 30, 70]
    plt.figure(figsize=(7, 4))
    bars = plt.bar(langs, users, color=["#4c72b0", "#dd8452", "#55a868", "#c44e52", "#8172b3"])
    plt.bar_label(bars)
    plt.title("Programming Language Popularity - Bar Chart")
    plt.xlabel("Language")
    plt.ylabel("Users (%)")
    plt.show()


exp_3b()
```

Output

```
(see graph below)
```

Graph



Experiment 3.c

Question: Conduct an experiment to show data visualization using pie chart.

Code

```
import json
import re
from collections import Counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

def exp_3c():
    labels = ["Python", "Java", "C++", "R", "SQL"]
    sizes = [35, 25, 15, 10, 15]
    plt.figure(figsize=(6, 6))
    plt.pie(sizes, labels=labels, autopct="%1.1f%%", startangle=90,
            explode=(0.08, 0, 0, 0, 0), shadow=True)
    plt.title("Language Share - Pie Chart")
    plt.show()


exp_3c()
```

Output

```
(see graph below)
```

Graph

